

Instructor Solution Script

Total Duration: 45–60 Minutes

Part 1: Assignment Overview (5 Minutes)

Hello everyone.

In this assignment, we will build a simple AI Product Card Generator using Gemini AI.

The user will enter a product idea, and Gemini will generate:

- Product Name
- Tagline
- Description

We'll then display the result inside a beautiful product card.

By the end of this assignment, you'll learn:

- How to call Gemini API
 - How to send prompts
 - How to display AI-generated content on a webpage
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Part 2: Project Setup (5 Minutes)

Create three files:

- index.html
- style.css
- script.js

Connect CSS and JavaScript files.

Run the project using Live Server.

Part 3: Building the UI (10 Minutes)

Create:

- Heading
- Input field
- Generate button
- Empty product card

Add basic styling:

- Center content
- Add shadows
- Rounded corners
- Good spacing

Show students how the UI looks before AI integration.

Part 4: Understanding the Prompt (5 Minutes)

Explain prompt engineering.

Example prompt:

Generate:

1. Product Name
2. Tagline
3. Short Description

for this product idea: "AI Study Assistant"

Return the response in JSON format.

Discuss why structured output makes parsing easier.

Part 5: Gemini API Integration (15 Minutes)

Steps:

1. Get Gemini API key
2. Store API key
3. Create async function
4. Send prompt to Gemini
5. Receive response
6. Convert response to JSON

Show students the complete API call.

Test with multiple product ideas.

Part 6: Displaying Results (10 Minutes)

Update:

- Product Name
- Tagline
- Description

inside the product card.

Demonstrate dynamic DOM manipulation.

Test:

- Smart Water Bottle
 - Travel Planner
 - Fitness Coach
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Part 7: Bonus Feature (5 Minutes)

Add one simple enhancement:

- Random emoji OR
- Copy button OR
- Color theme changer

Explain that bonus features are optional.

Part 8: Submission Instructions (2 Minutes)

Students can either:

Option 1: Paste prompt and code directly into the answer box.

Option 2: Upload files to Google Drive and share the Drive link.

Required submissions:

- Source Code
- Gemini Prompt
- Screenshot

Congratulations! You have built a complete AI-powered mini application using Gemini.